

UserJourney2Vector: Enterprise Application of Transformer Models on User Telemetry Data

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ABSTRACT

User telemetry data offers valuable insights into user behavior and goals when interacting with digital products and services. However, analyzing vast amounts of event sequence data can be challenging. In this case study, we apply transformer neural networks to telemetry event sequence data, yielding the UserJourney2Vector model, which treats event sequences similarly to natural language to distill complex user journeys into latent vector representations. The model enables understanding of typical and anomalous user paths, prediction of next actions, and user segmentation based on behavior. We used the model to obtain user clusters, predictive journeys, and cluster statistics, then conducted interviews with digital product experts to assess potential applications and impact of the model output. Experts responded positively to the model's ability to bolster user personas with data-driven insights and noted integrating model outputs with current practices could augment product design and data interpretation.

CCS CONCEPTS

• **Computing methodologies** → **Learning latent representations**; *Neural networks*; *Dimensionality reduction and manifold learning*; • **Human-centered computing** → **HCI design and evaluation methods**.

KEYWORDS

Design Methods, Quantitative Personas, User Journeys, User Studies, Representation Learning, Transformers

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1 INTRODUCTION

Understanding user behavior and intentions is a key element to human-centered design of digital products [13]. Data scientists, designers, and user experience (UX) researchers look toward vast amounts of event sequence & telemetry data, triggered by user

interactions, to gain insight into user actions and objectives. However, with complex enterprise applications and products, simple event plotting and even more advanced clustering and filtering methods often result in dense and difficult visuals to parse [7, 10], limiting conclusions to be drawn. Further, understanding and distilling complex user journeys with loops, errors, and a wide variety of objectives in a product is nearly impossible through linear event sequencing. This can result in designers and data scientists filtering and grouping events, sequences, and data that could introduce biases held in their product streams [4, 5]. If vast telemetry data can be parsed, filtered, and distilled effectively, it could unlock insights purely qualitative means cannot toward user pain-points, commonalities, and anomalies. This is especially potent in fast-paced business scenarios as new product features are implemented and user traffic increases, staying at pace with incoming data is imperative for effective design decisions.

1.1 Quantitative Methods for Persona Creation and User Journeys

Personas (fictional persons that represent real user segments) and User Journeys (activities/processes performed to achieve a certain goal) are utilized in UX research and design to determine relevant user processes, build empathy, and identify product areas with need for development [5, 13]. While robust qualitative methodologies of these principles exist, quantitative research and production leveraging event sequence data are increasing as well [3, 12]. Organizations use data from users traversing digital products to influence many spheres of product development (e.g., customer profiling, marketing, client acquisition, demand forecasting, predictive analytics, customer relationship management [3]). However, event sequence data can be unpredictable – sparse or dense, irregular or consistent, occur in parallel or strictly sequential. Still, it lends itself to many types of insights: finding common patterns, outliers, prediction, identification of predictive factors. Quantitative persona creation primarily uses clustering methods, followed by non-negative matrix factorization, with increasing methodological diversification over time such that practitioners can leverage user data to create shared mental models of user segments [11]. However, these methods generally focus on core/majority users while not taking advantage of insights on usability from outliers or minority segments that may appear over time and as features are added or changed in a product [13]. Quantitative methods of understanding and predicting user journeys in various contexts include visual analytics [7], Recurrent Neural Networks [9] (used to predict diseases based on the historical sequence of medical events), Variational Autoencoders [6] (to detect anomalous events by comparing sequences with expected normal progressions), and Markov Chains [1, 16]

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(probabilistic modeling focusing on short-term user behavior based on the current state). New research and advancements in machine learning could better leverage telemetry data for increased impact in user focused digital enterprise product design.

1.2 Present Study

With these principles and latest research in mind, we set out to present an application of transformers with extra-large context [8] on event sequence data to distill user behavior in digital systems. By capitalizing on current research on latent representation models for text, we can offer a powerful approach to learning complex underlying patterns of user behavior in vast amounts of event sequence data, allowing us to extract insights and surpassing the capabilities of traditional methods in terms of accuracy, expressiveness, and flexibility [15]. We offer our UserJourney2Vector model toward distilling user behavior and utilizing the generated output to segment users and evaluate and predict behavior for enterprise product design. Additionally, we present a formal case study evaluation to examine potential impact and opportunities for further application and development of this model with a variety of product design experts.

2 USERJOURNEY2VECTOR MODEL

The UserJourney2Vector Model is an application of Vaswani et al.'s [15] Transformer architecture for transformer embeddings on user telemetry data. The UserJourney2Vector system can take many forms, each of which comprise a sequence of event tokens to a transformer encoder that distills the sequence into a latent vector containing the information needed for a downstream task (Fig. 1). The decoder converts the latent representation vector back into a form usable for the downstream task. Thus, one may specify their desired task by constructing an appropriate decoder. For example, if the downstream task is predicting the next user action, a feed-forward neural network outputting event probabilities makes a suitable candidate for the decoder. In this case, the latent vector represents the mutual information between all the previous tokens and the next token. i.e. the vector excludes any information not useful in predicting the following token. For learning persona clusters, we desire a single vector that represents as much information about the journey as possible. Then, affine transformations, as in the contrastive representation learning approach, will suit the task. Ultimately, the chosen task plays a secondary role, simply controlling the information contained in the latent vector, which we primarily concern ourselves with. The flexibility of the model with respect to different downstream tasks allows for a wide range of distillations that highlight different insights about user behavior. See Table 1 for potential use cases given model output. We demonstrate two tasks, prediction and persona clustering, as examples for distilling useful information for stakeholders involved in building product user experience. We begin here by first introducing the structure of the model. Up-to-date code for the model is available at <https://github.com/dycontri/uj2vec>.

2.1 Transformer Encoder

The transformer encoder plays a key role in distilling the user journey sequences. Similar to a series of words in natural language data,

Table 1: Potential uses of prediction and representation model types

Model Output Type	Potential Use Cases
Prediction	Next page visit Next product purchase Provide relevant recommendations Provide personalized content Product recommendations Marketing Campaigns to targeted users Identify at risk to churn users Predict demand for products and services Improve inventory management
Representation	Understand user journey Identify patterns in behavior Detect fraud or anomalies in behavior Segment users Improve recommendation systems

events that occur in user journeys exhibit sequential dependencies. Just as the English word "bad" takes on different meaning when preceded by the word "not," events such as "End" change meaning when preceded by "Error" or "Success." As in models of natural language, transformers' attention mechanism allows the distilled representations to contain information about these sequential dependencies. For user journeys with thousands of events, it may be necessary to divide sequences into multiple, smaller segments. Then, extra-large attention contexts, as in [2], retain information across the segments.

2.2 Latent representation and Decoder

The decoder controls how the model converts the abstract latent representations into concrete data used for the downstream task. For predicting future user journey events, a feed-forward neural network decoder converts the latent representation directly into a probability distribution of future events through a non-linear transformation. For learning user persona clusters, a linear transformation projects the latent representation into a space describing the mutual information between views across the entire user journey, as described in [14]. Then, clusters in the user journey representations can be identified using whatever clustering technique desired, including parametric methods such as K-means or non-parametric methods like Dirichlet Process Mixtures.

3 CASE STUDY EVALUATION

To assess the business impact, potential uses, and gaps of the output of the UserJourney2Vector model we gathered various experts on digital product design and telemetry data for feedback. For our case study we chose two areas of application of the model output for evaluation and feedback: (1) user clusters, including Net Promoter Score (NPS) of the user clusters, and (2) predictability of a user journey, with implicit output of prediction of next step of a user journey. See Section 4 for further discussion on the various applications and uses of the output of the UserJourney2Vector model.

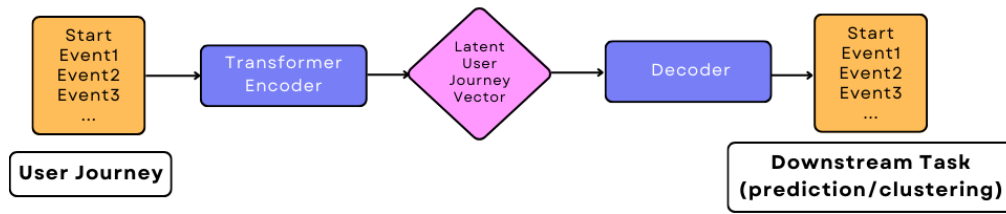


Figure 1: Schematic diagram of the model, the latent vector contains information relevant to the chosen downstream task

3.1 Methods

Individual interviews were conducted via Microsoft Teams and lasted between 40 minutes – 1 hour, all were conducted in September 2023. We chose a variety of experts that touch differing parts of the product design and evaluation process for feedback. Interview participants consisted of front-end (2 interaction designers) and back-end (2 data scientists and 2 product managers) practitioners: all held expertise crucial for product development and user experience. Participants were familiar with digital product telemetry data and had some experience with AI/ML evaluation and design of enterprise products. The interview transcripts and recordings were analyzed using an inductive approach.

3.1.1 Interview Protocol. The aim of the semi-structured interviews was described to experts as helping us gauge how the UserJourney2Vector model and output visualizations could aid and supplement decision making in product design and development. Interviews began with opening questions around current usage and gaps of personas and user journey mapping in their space (e.g., *How do you currently think about and utilize Personas/User Journeys in your work? What are the benefits/limitations of this framework?*). Following this, we prompted experts to share how telemetry data is currently used in their product teams and offered 3 common event sequence visualizations for feedback: an arc diagram, a Sankey diagram, and a sunburst diagram (see Fig. 2 for examples). We asked experts if the presented visual was familiar to them and the usefulness and limitations of the chart. We chose aggregated event sequence data charts and raw sequence data charts for various insights and scenarios. Following this, we discussed the UserJourney2Vector model and its capacities: ability to consider the full length & sequence of user interactions and significance of particular events, comprehension of a typical and unique user journey, prediction of a user’s next action, and categorization of users based on their objectives and interactions with the product. We asked for initial thoughts and then displayed some applications of the model and its outputs for demonstrative purposes. While many visualizations could be created with model output, we chose three that represented user segmentation and prediction to illustrate potential uses of the model for our experts. In practice, we expect practitioners would choose visualizations familiar to their teams and useful to their purposes. We chose to display the user clusters chart followed by the clusters’ NPS as an example statistic that our experts were familiar with (Fig. 3). We asked experts how these outputs could be useful and what other insights or applications they would like to see. We ended with a visualization of the likelihood of a single user’s journey (Fig. 4). We used data from an application experts were

familiar with and described how the model created the clusters and journey map visualizations. We allowed experts to ask clarifying questions/offer feedback about the visualizations themselves, but primarily used the visualizations as a tool to help experts understand and imagine potential outputs and uses of the model. We also discussed the application of the model’s ability to predict a user’s next step and detect anomalies in user journeys. Closing questions included prompting on the initial event sequence visualizations and their relation to the model output, potential impact and usability of model output, as well as gaps and improvements/changes to be made. Full semi-structured interview guide is available in the supplementary material.

3.2 Findings

In this section we discuss the key themes that emerged from our analysis. To protect anonymity, participants are referred to by using ‘PM’ for product managers, ‘D’ for designer and ‘DS’ for data scientists, followed by a participant number.

3.2.1 Current Usage of Personas and User Journey Mapping. Both our product managers (PMs) and designers discussed the frequent usage of established personas within their product groups. Formal personas are being phased out in some product groups as they turn to ‘User Roles.’ User Roles describe a job role in the product, rather than a person, for inclusivity as organizations vary considerably on how they assign people to specific jobs; a User Role may have more flexibility than a specific persona. User Roles are then connected to workflows – differing User Roles may be responsible for different jobs within a larger workflow. They come together to create an optimized ecosystem of touchpoints that support all the steps and tasks for a cohesive end-to-end experience for jobs and journeys across User Roles. D1 described User Roles as a “guiding star on how we go about solving particular problems” while for D2 the main benefit was “a shared understanding who you’re designing for and why.” Our data scientists, however, did not mention use of User Roles in their daily work, but that their product teams might have a commonly discussed “key golden path” [DS2]. Experts noted the limitations of User Roles however as it is important to “keep in mind that they are oversimplifications of the actual user base of your product” [D2].

“The limitation is how often do we revise those User Roles and how detailed are they? How frequently are they changing and are we up to speed as we approach more users? Are our Personas evolving? Let’s say we get to 3 times, 4 times million more users, do our assumptions still hold, I don’t know? It would be great to know.”

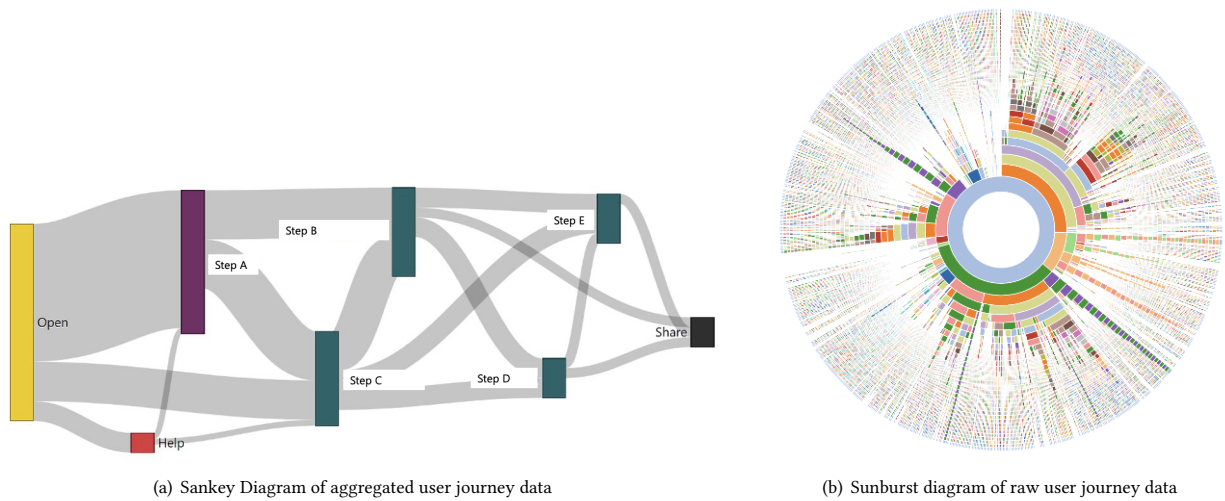


Figure 2: Examples of typical visual analytics utilized for user journey telemetry data

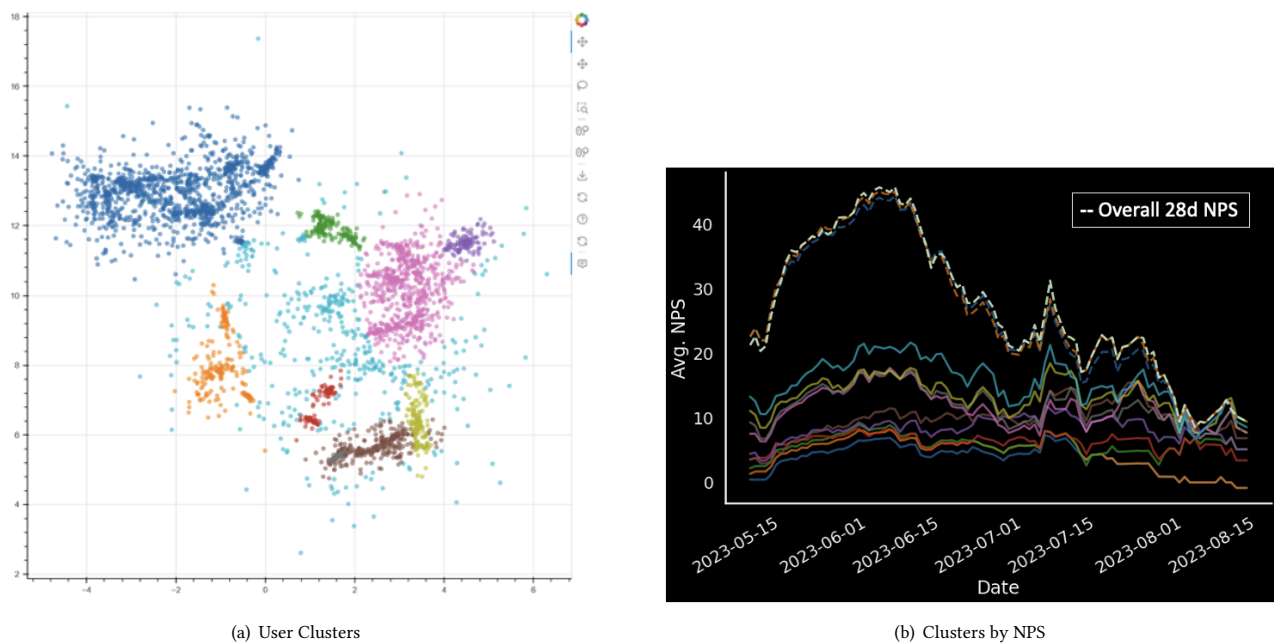


Figure 3: Example output visuals offered to experts. Visuals were chosen to illustrate potential uses of the UserJourney2Vector model that were familiar to experts. Practitioners could use visualizations similar to these or ones more suited to their needs and audience.

[PM1]

“There might be 10-20 people interviewed as we try to cover User Roles. It would give us more confidence in our assumptions and on features that we are building to say, hey, this is the data backing it up” [PM2]

“The limitation can be sometimes that we only have a couple of them defined, I have had instances where there

was a User Role that wasn’t fully defined yet, but I knew that that was the persona that needed to be attached to the certain part of the user journey. So I had to kinda make it up a little bit based on what I thought about a loose persona.” [D1]

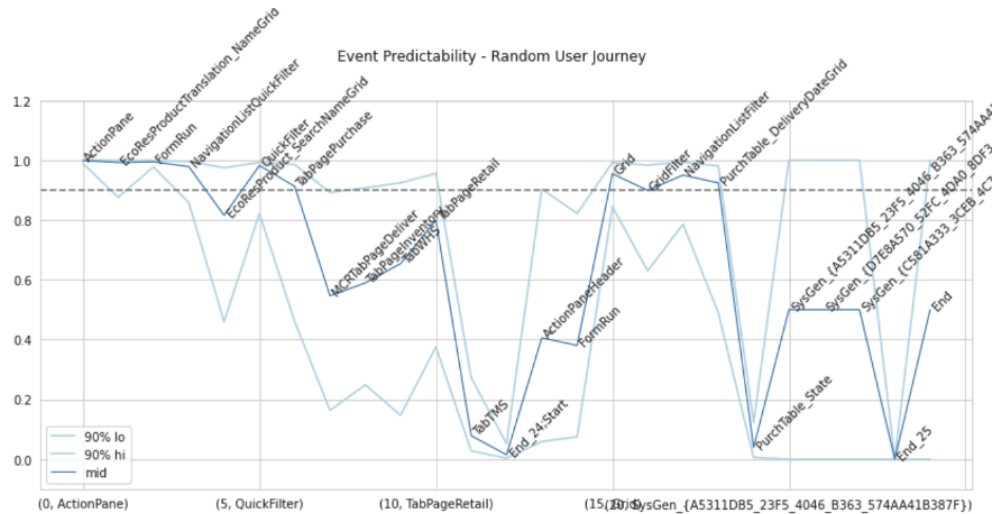


Figure 4: Output visual of likelihood of user steps in a random user journey with 90% confidence intervals used for illustrative purposes

3.2.2 Event Sequence Visual Analytics. Even before displaying the visualizations, the data scientists commented on the difficulty of parsing vast amounts of often messy event sequence data for their product teams to digest.

“The main issue is that it’s very hard to apply journey visuals to product telemetry data without doing a bunch of aggregations. Once you do the aggregations you get a pretty graph that isn’t telling you very much so you get a trade off between those two...to get the most we just use statistics on what percentage of people do X activity, it’s mostly calculating aggregated statistics.” [DS1]

“We’ve done Sankey diagrams and Force-Directed charts, but the challenging part is there’s always a really long tail of permutations of journeys people can take.” [DS2]

As there is no way to tie demographic information or User Role directly to incoming telemetry data, event sequence visualization is often used to display data. Most participants were familiar with aggregated sequence visualizations and were comfortable interpreting them, as they are “tangible” [D2] and “similar to funnels following users as they navigate through the product” [PM2]. Experts spoke about wanting “more details” [D1, PM1, PM2] on specific steps within the larger groups while mentioning it would be “helpful to know about the users” [D1, PM1] and if certain “cohorts” [PM1] were going down certain paths. DS2 offered a specific pain-point that “we always tried to validate, how important is the order versus just the fact that this event happens.” Further, teams often have “a fundamental assumption” that a certain order is correct, however “we don’t even know the order people want to go in, it’s just missing” when event sequence data is simply aggregated [DS2].

The sunburst diagram (Fig. 2(b)) displays raw event sequence data, but is limited to 30 steps. This offers a more “granular” [DS2] view of users, helpful for “drilling down into the data little better” [DS1]. The sunburst diagram was interactive and experts were shown the steps and percentage of users that went down a path as

they mouse-over the rays of the diagram. It was noted that some paths were extremely similar (with even 1 step being different) but not necessarily grouped together in the sunburst. All experts noted the sunburst is “visually overwhelming” [D1, PM1, PM2] or over-crowded and a “limited subset” [DS1, DS2] of users would be more useful to dive into.

3.2.3 Model Output, Applications, and Impact. Experts commented on the simplicity and usefulness of user clusters in their work and how event sequence data can have higher impact on design by capitalizing on user clusters. All experts shared the sentiment that user clusters “coming from telemetry is a great opportunity for quant/qual research – it could give us new users that researchers are not thinking about that are actually being accounted for in the model” [DS1]. Experts noted NPS is a helpful statistic to see split by clusters, along with many other metrics over time to “help us focalize where the issues are” [PM1, D1].

“So this is exactly what I want have in my day-to-day work...I’m not going to look at these day-to-day, but every quarter, is the persona the same? I’m introducing new work, is it helping, is it changing, is it affecting our clusters? It’s the live data I’m needing.” [PM1]

“Within this classification we can figure out if people are using the product differently...there could be a new emerging pattern that we didn’t know about it,” [PM2]

“It’s always the dream of tying a qualitative persona to telemetry, it’s just always really tricky. This might be useful for approaching it in the reverse based on what people are doing.” [DS2]

Experts commented that prediction of user journeys would aid in “understanding what the edge case is and how detrimental would it be if people continue to fall into edge cases” [D1] while also considering golden paths the model could point to: “what percentage of our journeys are highly volatile and which are really predictable and does it align with the highest volume paths?” [DS2]. Both PMs

discussed using the prediction model to “catch drops” and actions “not expected” by the model to prevent potential issues with the user interface.

Importantly, all experts acknowledged the application of automation of high volume tasks and potential to help users move through pain-points: “we can navigate them through to a success point and know which actions to automate” [PM2]. Potential limitations of interpretability and broader “accuracy and confidence concerns” [PM2] of the model at large were discussed as well: “Let’s say the model shows 3 clusters, what if it’s the same kind of persona documented? What does that mean?” [PM1]. DS2 wondered, “how distinct the clusters are, just because it would depend on the product – is it an individual doing things that leads to an outcome or a team with shared journeys that will lead to a certain outcome?” Experts were also interested the potential and future applications of the model output such as aiding in understanding “time to value for a user” [PM1].

“I would ask a data scientist, in what instances do these clusters overlap, what would prompt that? Would it be a certain path, or at what point in their flow is there a failure? Is there a space we’re not getting them through the flow fast enough?...What can we do to improve the experience of these new clusters? How do we work them into the product? Maybe we didn’t educate a certain cluster enough about a feature that pertains to them or re-design for them.” [D1]

“Can we ask the LLM to summarize the similarities of the groups? For the predictability of paths, I’d want to know how that fits into the larger context. Even if we found 6-8 personas with distinct paths, but then when we look further, paths were similar between groups. Or maybe we treat everyone the same and then at X point we can treat them differently” [DS2]

Experts advised that model output is best used in concert with qualitative measures as this output could “lend itself to understand personas that aren’t fleshed out and figure out who they are and jobs to be done” [D1]; “pairing [model output] with usability studies, user interviews, and actual conversations would really help a team or a designer get a holistic view of the problem they’re trying to solve and impact the way they would approach it” [D2]. Experts also noted the power of combining model output with the event sequence visualizations as they can limit and compare data across clusters or specific journey types: “I see them as complimentary. You could have a sunburst or network that is super useful at an aggregate, and a filter for different clusters” [D1].

4 DISCUSSION & CONCLUSION

There are many approaches to parsing vast amounts telemetry data of enterprise products to understand user journeys and behavior. The UserJourney2Vector model we offer here can not only aid in this task but increase effectiveness of current methodologies such as visual analytics for user journeys, persona clustering, and predicting user behavior. This architecture is more efficient than probabilistic methods as it can capture long-range dependencies and model a wider range of data types and distributions. Additionally, the UserJourney2Vector model is adaptable to new tasks and domains;

it can understand complex relationships between variables (e.g., user interests, goals, context of interaction), generate new data for simulation or testing, and contains a lower-dimensional representation of the telemetry data, lending itself to further visualization and machine learning tasks. The latent representation within the UserJourney2Vector model holds a wealth of information that can be capitalized on: interests and goals inferred from pages clicked and order visited, navigation path, time on each page, interaction with pages, level of engagement, potential user satisfaction. It can be used to predict future outcomes of users and could be applied toward improving user experience through relevant recommendations, marketing, identifying users at risk of churn, predicting demand, or simulation of user behavior. It can further be used toward gaining deeper understanding of user journeys and patterns, detecting fraud and anomalies, understanding significance of certain events in a specific sequence, segmenting users based on behavior, improving recommendation systems, and more. While the UserJourney2Vector model can be computationally complex and it can be difficult to interpret transformer models, we believe this model offers an impactful application toward understanding user behavior in digital products.

We have made the code to the UserJourney2Vector model publicly available at <https://github.com/dycontris/uj2vec>. Further, we conducted a case study interviewing 6 experts in the digital product creation process (product managers, UX designers, and data scientists) to assess the usability, understandability, and potential applications of the UserJourney2Vector model. We gained an understanding of how they currently conceive of the benefits and drawbacks of Personas and user journeys, how user telemetry data is currently used, and offered traditional visual analytics of user telemetry data for feedback. We then introduced the UserJourney2Vector model and assessed the impact it might have on the use and conception of Personas, user journeys, visual analytics, and other methods of digesting user telemetry data.

4.1 Reflection and Lessons Learned

Our case study highlighted many facets of product design and broader applications for researchers. We found that gaining insight on complex user journeys using telemetry data is difficult with visualization alone. We saw a need for parsing vast data to make it human-digestible while being able to take advantage of outliers, anomalies, patterns, and distinct user behaviors. We observed that our model output was best used toward supplementing and supporting current methods of understanding and visualizing user telemetry data. Our interviews demonstrated that it is important for data scientists and researchers to fully understand the needs, goals, and common tasks of potential audiences to decide how to display and communicate model output.

As our interviewees varied in expertise, their touchpoints and experiences around digital product creation were different. User Roles were more present in the daily work of designers, quarterly work of product managers, and uncommonly referred to by data scientists. The benefits of utilizing User Roles and connecting them to journeys are apparent in their space, but the drawbacks of small sample sizes, revising, and uncertainty around inclusion of all users

was noted. Designers and product managers were more comfortable with their typical user journey charts (Fig. 2) and sharing their thoughts/general takeaways on user clusters broken by NPS (Fig. 3(b)) and the likelihood of user steps (Fig. 4) as the information was more useful for their daily goals. They preferred familiar visualizations broken down by the model output such that actionable conclusions could be achieved. Our data scientists focused on current user telemetry data, they discussed the messiness of raw data and the trade-off with aggregation. While getting data in a state of readiness for modeling is a major challenge, once there, one can be limited by the interpretability and variability of machine learning methods for stakeholders. Further, biases around how a product should be traversed, or which User Role does what activity can affect outcomes. Data scientists tended to focus on the statistical processes, challenges, and uses behind model output (e.g., user clusters (Fig. 3(a)) and journey prediction) rather than potential conclusions from the charts displayed. The visual analytics presented reinforced these sentiments and that visualizations are used, preferred, and interpreted differently based on expertise.

The potential impact of the UserJourney2Vector model on interpretation of user telemetry data was provocative. Experts felt the model output could challenge and bolster qualitative methods and they could integrate the output and its various applications to their daily work. The output could directly affect product design through automation and recommendations while offering insight into undiscovered user behaviors, help with pain-points, and augment current methods of parsing telemetry data. Utilizing the UserJourney2Vector model in concert with other methodologies was exciting to our experts – the segmentation can be clustered and run by indicating statistics, recommendation systems can be created, and visual analytics of the data could be utilized to greater impact. Further, machine-bias and human-bias can be reduced in building User Roles and key user journeys when pairing model output with qualitative measures. The UserJourney2Vector model has a unique ability to understand and capitalize on anomalies which could lead to more inclusive data interpretation and product design. Experts noted that integrating the UserJourney2Vector model requires clean telemetry data and product/data expertise to compare model output to expectations.

4.2 Outlook

We present the UserJourney2Vector model as an application of a generative transformer model structure that distills and capitalizes on user telemetry data in enterprise digital products. In future versions of the UserJourney2Vector model we hope to develop new ways of evaluating the quality of its latent representations as well as identify additional downstream tasks that allow the model to distill diverse types of information, such as predicting aspects of the user experience, like satisfaction or difficulty of use. Our goal is to help UX researchers, designers, producers, and data scientists take advantage of vast amounts of telemetry data and harness it toward their creative goals. Integrating a model architecture like this into current quantitative and qualitative methods of parsing user data, interpreting user behavior, creating personas, and understanding user journeys, could enhance and improve the entire product design process.

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